# SERA ERTUĞ



## PERSONAL INFORMATION


seraertug@gmail.com

+90 537 491 00 53

İzmir/Türkiye

www.linkedin.com/in/sera-ertug/

https://github.com/seraertug


## PROFILE

I build interdisciplinary projects that apply data and AI to engineering and the sciences. My goal is to turn research findings into real products that help people.

## EDUCATION

**Tevfik Fikret Science High School, İzmir (Science-Mathematics track)** 2023 - 2027
**Full (100%) merit scholarship**

**Tevfik Fikret Middle School, İzmir (Valedictorian)** 2019 - 2023

**Tevfik Fikret Primary School, İzmir** 2015 - 2019

**BİLSEM (Science and Art Center for gifted students)**
Selected in primary school through a two-stage national process: a group tablet test (visual-spatial intelligence), then an individual IQ assessment. Project track in science and technology in high school.

## TESTS

- AP Calculus BC (5/5)
- AP Physics 1 (5/5)
- Sir Isaac Newton Physics Contest (7th worldwide / 2,274 participants; 1st in Türkiye / 196)

## LANGUAGES

- Turkish – Native
- French – Advanced (C1, DALF)
- English – Advanced (IELTS: November 2026)

## INDEPENDENT RESEARCH

- Effects of extreme temperature changes in childhood on adult brain connectivity
- Fractal texture of abdominal fat as a potential biomarker for PCOS

## SCIENCE & TECHNOLOGY

**ETH Zurich AI Summer Week (65 selected, 23 countries)** 2026
Built an AI-based business model with an international team and pitched it to a jury including Tesla and Google representatives. Honorable Mention

**Beamline for Schools Competition, organized by CERN , Team Phoenix** 2025
Project "Invisible Probes of Marble": muon scattering and AI-based data analysis

**Junior Software Engineering Intern - VESTEL**

**Conrad Challenge - Conrad Innovator**
Designed a tooth-mounted micro-sensor that tracks metabolic health in real time through saliva analysis

**CS50x: Introduction to Computer Science (Harvard University, via edX)** 2024
10 problem sets + 1 final project

**Sigma STEAM Olympics - 1st Place**
Cosmic Acoustics: software that turns galaxy images into music using pixel analysis and the golden ratio

**Robotex International - Finalist, Software Technologies category, Team Phoenix**

- International Finals, Tallinn, Estonia
- National Finals, 3rd place, Antalya, Türkiye
- Aegean Regional Finals, 2nd place, İzmir, Türkiye

**Inspirit AI Scholars - Online Summer Program for High School Students**
Interdisciplinary final project: AI-based sustainable agriculture

**AI Marathon & Hackathon by Turkcell and the Ministry of National Education**
National Hackathon Finalist, Jury Special Award

## SOCIAL IMPACT

**Youth Sustainable Development Leader - SÜGEP** 2026

- Technical bill on photovoltaic systems at the Grand National Assembly of Türkiye
- Public awareness campaign (5,000+ signatures)
- From technical data to policy, through a field study of a LEED Platinum-certified building

**Speaker and Volunteer - Global Sustainable Development Congress, Istanbul, Türkiye** 2025

**Youth Climate Essay Contest (Climate Cardinals & Yale Climate Connections) - Top 10 submission**

**Speaker at GSDC as a member of the SÜGEP team - Bangkok, Thailand** 2024
Presented our project "A Model Carbon Capture Project"

## ARTS & SPORTS

- ABRSM Piano Exam (Grade 5) – Passed with Merit
- Mozart Academy IV International Piano Festival – Outstanding Performance Award
- Rhythmic gymnastics (ages 6–12): national and international awards
- Tennis: tournament player